TASK 4: MIXING RATIOS, VISCOSITY, VOC AND MATERIAL PLANNING

Vehicle Damage Paint Technician Level 3 | Block 4 | v1.1

TIME	EVIDENCE	PROGRAMME POINT
60 minutes	**Practical | Knowledge | Documentation**	**Weeks 43-56**

Your task

A refinishing job requires materials to be planned and mixed accurately. You must calculate material needs, confirm ratios, consider viscosity and waste, and record how the mix supports compliance and quality.

What you must do

1. Confirm the repair area, product system and stage of refinish work.
2. Use product data to identify ratio, hardener/thinner options and pot life.
3. Calculate the estimated quantity needed and record any waste-control decisions.
4. Prepare or simulate the mix using the correct measuring method and controls.
5. Record the mix data, equipment used and any issues found during planning.

What good evidence looks like

SAFE SETUP

Suitable PPE, a prepared work area and controlled vehicle, component or equipment.

CORRECT METHOD

A logical sequence using the relevant procedure and technical information.

ACCURATE EVIDENCE

Actual readings, findings and actions recorded clearly and precisely.

SOUND DECISIONS

Appropriate recommendations, referral or authorisation supported by evidence.

You will need

- Paint product data sheets and mixing scheme information
- Mixing cups, scales, viscosity cup where available and stirrers
- Sample formula or simulated mixing materials
- VOC/waste control information and PPE
- Material planning record

SOW ALIGNMENT

Block 4 - Colour Theory, Paint Mixing and Technical Data
Duties: D4, D5, D7, D10 KSB evidence focus: K1, K3, K6, K8, S3-S5, B4, B7
Task reference: Block_4_Task_4_Mixing_Ratios_Viscosity_VOC_and_Material_Planning

TASK 4: MIXING RATIOS, VISCOSITY, VOC AND MATERIAL PLANNING

Practical record

Apprentice, location and vehicle details

Apprentice name

Date

TASK COMPLETED AT ○ APPRENTICE'S WORKPLACE ○ TRAINING CENTRE

Vehicle make

Model

Registration

Mileage

Panel / repair area

VIN / chassis number

Measurements and technical checks

Product selected		Mixing ratio	
Hardener/thinner		Pot life	
Quantity planned		Viscosity check	
VOC/waste control		Mix evidence reference	

Formula, product data, mixing calculation, viscosity check, waste or VOC references

Task evidence

PPE and precautions used

Safety-critical findings and immediate action

Work completed, findings and recommendations

Knowledge check

1. Why must paint materials be mixed using the product data sheet?

2. What could happen if too much hardener or thinner is added?

Knowledge check - continued

3. (a) Why is pot life important when planning material use? (b) How does accurate quantity planning reduce waste?

4. What is viscosity and why may it matter during application?

5. (a) How can VOC and waste controls support environmental compliance? (b) What records should be kept for material traceability?

Assessor/Workplace Mentor observation

- ☐ Interpret product data for ratios, hardeners, thinners and pot life.
- ☐ Record mixing, viscosity and VOC/waste considerations accurately.
- ☐ Explain how incorrect mixing affects finish quality and compliance.
- ☐ Calculate material quantity suitable for the repair area and process stage.
- ☐ Select suitable equipment and controls for safe material preparation.

Assessor/Workplace Mentor observation notes

Decision and feedback

Decision - select one

◯ **PASS** ◯ **REFER**

Evidence reference

Feedback and action required

Sign-off

Signatures confirm that this is the apprentice's own work and that the evidence and decision are accurate.

Apprentice name

Assessor/Workplace Mentor name

Apprentice signature

Assessor/Workplace Mentor signature

PIN (if applicable)

Date

Vehicle Damage Paint L3

Task answer continuation | Form accessibility revision 12 November 2026

Use this page when an answer will not fit visibly in its original box.
Reference the question and write "See continuation page" in the original answer.
Keep each response within the visible space. Add another copy of this page if needed.

Apprentice name

Block / task / page number

Question / section reference

Continued answer

Question / section reference

Continued answer

Continuation evidence must be reviewed with the original task and referenced at sign-off.